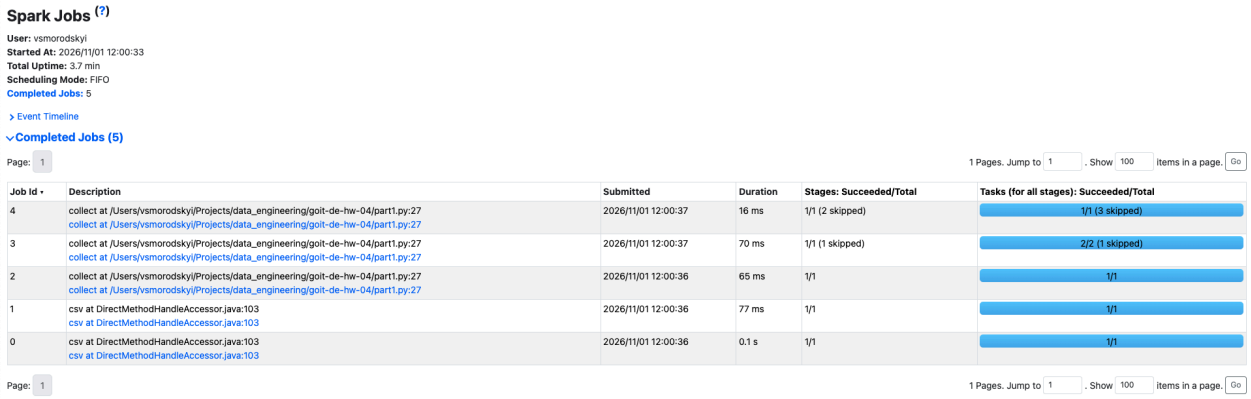
Частина 1



Spark Jobs (?)

User: vsmorodskyi
Started At: 2026/11/01 12:00:33
Total Uptime: 3.7 min
Scheduling Mode: FIFO
Completed Jobs: 5

Event Timeline

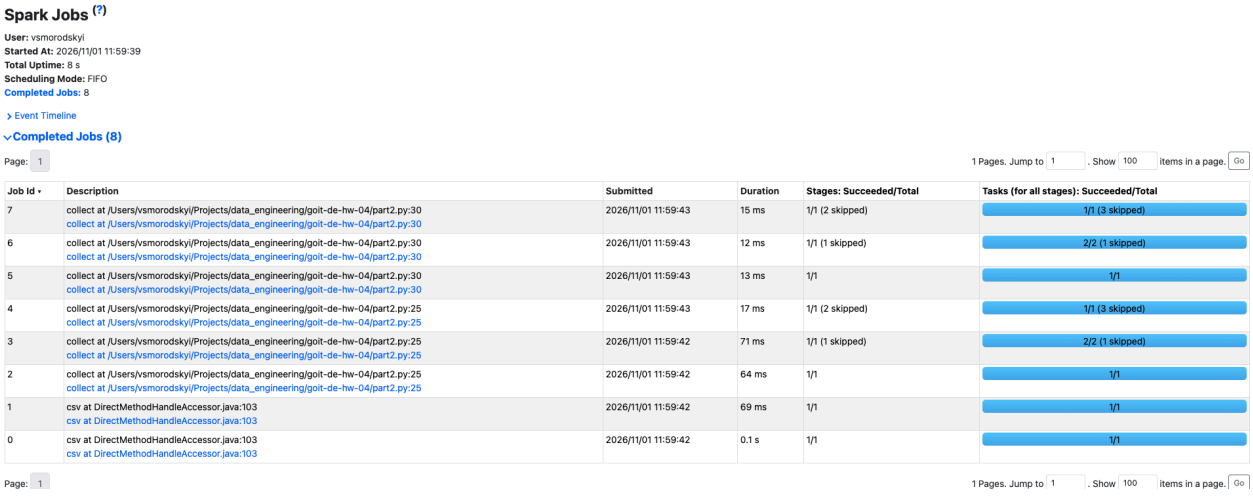
Completed Jobs (5)

Page: 1 — 1 Pages. Jump to 1 . Show 100 items in a page. Go

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
4	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part1.py:27 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part1.py:27	2026/11/01 12:00:37	16 ms	1/1 (2 skipped)	1/1 (3 skipped)
3	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part1.py:27 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part1.py:27	2026/11/01 12:00:37	70 ms	1/1 (1 skipped)	2/2 (1 skipped)
2	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part1.py:27 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part1.py:27	2026/11/01 12:00:36	65 ms	1/1	1/1
1	csv at DirectMethodHandleAccessor.java:103 csv at DirectMethodHandleAccessor.java:103	2026/11/01 12:00:36	77 ms	1/1	1/1
0	csv at DirectMethodHandleAccessor.java:103 csv at DirectMethodHandleAccessor.java:103	2026/11/01 12:00:36	0.1 s	1/1	1/1

Page: 1 — 1 Pages. Jump to 1 . Show 100 items in a page. Go

Частина 2



Spark Jobs (?)

User: vsmorodskyi
Started At: 2026/11/01 11:59:39
Total Uptime: 8 s
Scheduling Mode: FIFO
Completed Jobs: 8

Event Timeline

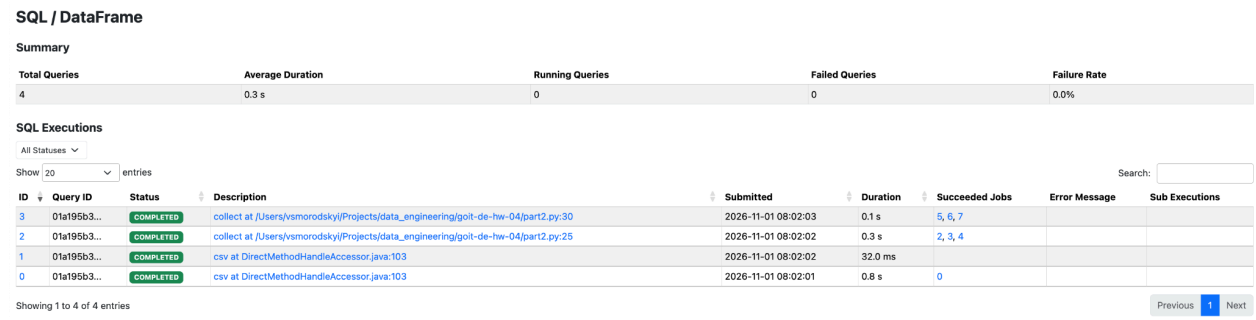
Completed Jobs (8)

Page: 1 — 1 Pages. Jump to 1 . Show 100 items in a page. Go

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
7	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:30 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:30	2026/11/01 11:59:43	15 ms	1/1 (2 skipped)	1/1 (3 skipped)
6	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:30 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:30	2026/11/01 11:59:43	12 ms	1/1 (1 skipped)	2/2 (1 skipped)
5	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:30 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:30	2026/11/01 11:59:43	13 ms	1/1	1/1
4	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:25 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:25	2026/11/01 11:59:43	17 ms	1/1 (2 skipped)	1/1 (3 skipped)
3	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:25 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:25	2026/11/01 11:59:42	71 ms	1/1 (1 skipped)	2/2 (1 skipped)
2	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:25 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:25	2026/11/01 11:59:42	64 ms	1/1	1/1
1	csv at DirectMethodHandleAccessor.java:103 csv at DirectMethodHandleAccessor.java:103	2026/11/01 11:59:42	69 ms	1/1	1/1
0	csv at DirectMethodHandleAccessor.java:103 csv at DirectMethodHandleAccessor.java:103	2026/11/01 11:59:42	0.1 s	1/1	1/1

Page: 1 — 1 Pages. Jump to 1 . Show 100 items in a page. Go

При додаванні ще проміжної дії collect() додаються ще три завдання через повторне виконання всього плану. Можна це перевірити на вкладці SQL/Data Frame. Там можна побачити айді завдань (jobs) буде той самий граф виконання (+ фільтрація, що додана на останньому етапі).



SQL / DataFrame

Summary

Total Queries	Average Duration	Running Queries	Failed Queries	Failure Rate
4	0.3 s	0	0	0.0%

SQL Executions

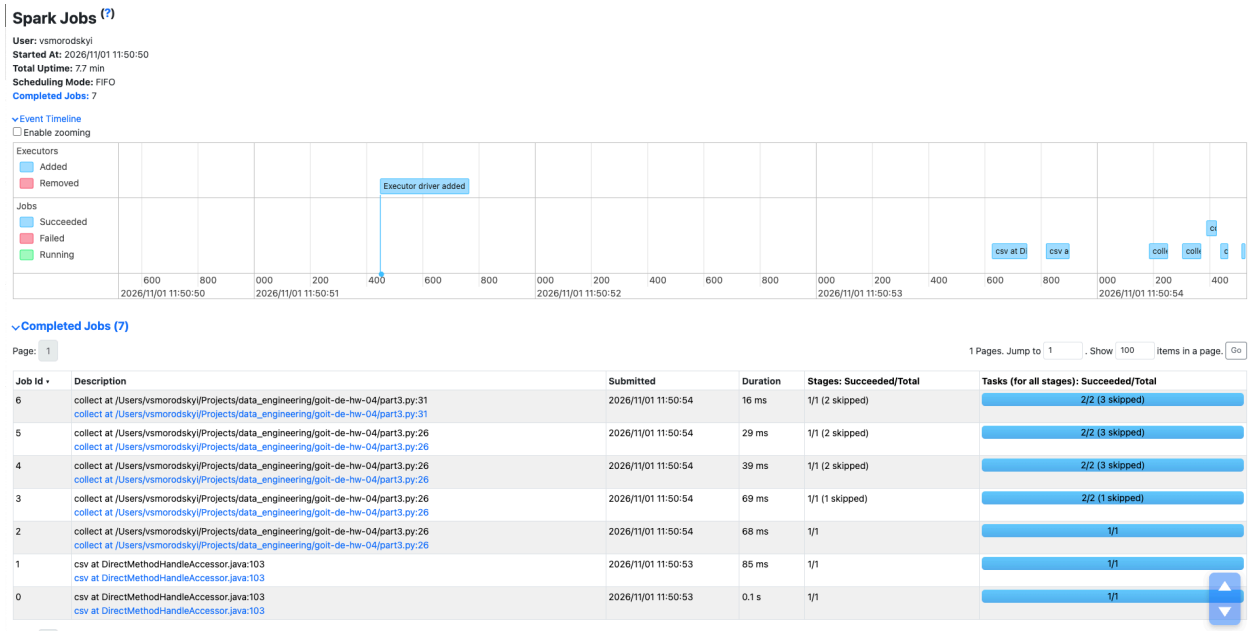
All Statuses

Show 20 entries — Search:

ID	Query ID	Status	Description	Submitted	Duration	Succeeded Jobs	Error Message	Sub Executions
3	01a195b3...	COMPLETED	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:30	2026-11-01 08:02:03	0.1 s	5, 6, 7		
2	01a195b3...	COMPLETED	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part2.py:25	2026-11-01 08:02:02	0.3 s	2, 3, 4		
1	01a195b3...	COMPLETED	csv at DirectMethodHandleAccessor.java:103	2026-11-01 08:02:02	32.0 ms			
0	01a195b3...	COMPLETED	csv at DirectMethodHandleAccessor.java:103	2026-11-01 08:02:01	0.8 s	0		

Showing 1 to 4 of 4 entries — Previous 1 Next

Частина 3



Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
6	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:31 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:31	2026/11/01 11:50:54	16 ms	1/1 (2 skipped)	2/2 (3 skipped)
5	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:26 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:26	2026/11/01 11:50:54	29 ms	1/1 (2 skipped)	2/2 (3 skipped)
4	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:26 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:26	2026/11/01 11:50:54	39 ms	1/1 (2 skipped)	2/2 (3 skipped)
3	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:26 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:26	2026/11/01 11:50:54	69 ms	1/1 (1 skipped)	2/2 (1 skipped)
2	collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:26 collect at /Users/vsmorodskyi/Projects/data_engineering/goit-de-hw-04/part3.py:26	2026/11/01 11:50:54	68 ms	1/1	1/1
1	csv at DirectMethodHandleAccessor.java:103 csv at DirectMethodHandleAccessor.java:103	2026/11/01 11:50:53	85 ms	1/1	1/1
0	csv at DirectMethodHandleAccessor.java:103 csv at DirectMethodHandleAccessor.java:103	2026/11/01 11:50:53	0.1 s	1/1	1/1

При використанні методу cache() кількість задач була зменшена за рахунок повторно використаних обчислень, що були збережені в оперативній пам’яті.